

Development of Training Systems using Spatial Augmented Hand

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ABSTRACT

Training to execute procedural tasks is often a burden for novice workers. Different systems have been developed to improve users' experience and, consequentially, their performance at a task. However, systems for tasks requiring hand operation lack fine details regarding a specialist choice of hand movements or positions. This study proposes a projection-based hand visualization system that guides the user's physical hand through a task.

Index Terms: Computing methodologies—Computer graphics—Graphics systems and interfaces—Mixed / augmented reality;

1 INTRODUCTION

Augmented Reality (AR) can support task performance and relieve the mental workload during procedural tasks. Compared to traditional instruction methods, such as paper-based training or an interactive display, AR systems can lead to shorter task completion time, reduced error, and better user awareness [2–4]. The operation differences between AR and traditional methods, such as requiring the complete or partial halt of the step in progress to operate the instruction manual or an interactive display, or the type of information it can provide, are the leading causes of better AR systems results [3].

Paper-based instruction usually does not provide enough information regarding hand movements, even when using diagrams and pictures. Seeing the actual task in action is crucial to understanding and proper execution. Interactive displays can contravene some problems caused by paper-based instruction. However, in many cases, the display operation will interfere with two-hands tasks, increase task completion speed, and increase task errors [1].

Direct guidance is another training method that can also be used, but it also poses problems as it requires intense supervision by qualified personnel, which is costly and not scalable. Therefore, alternative instruction systems, such as virtual or augmented reality training systems, are needed.

Some studies find that head-mounted display performance is worse than spatial augmented reality systems and even increases the mental workload compared to paper instructions or interactive digital instructions on a tablet. Head-mounted displays introduce other restrictions, such as low resolution and poor field of view, negatively impacting the user's performance as they have to compensate for these deficits. Projection-based systems, on the other hand, do not show these drawbacks, performing better when compared to more traditional approaches.

Ueda et al. (2017) investigated the effectiveness of a hand projection technique, where they extend the user's arm and hand through projection to reach far away objects. Their results show that projection-based visualization of the hand is helpful in human-human communication as the user can convey information through



Figure 1: Our method projected hand of the trainer on the trainee's side to instruct the trainee

gestures by the projected hand. However, the current projected hand system assumes an appropriate projection surface around the user [5].

For a case where no suitable projection surface is available, we propose a projection-based system to support training situations where the focus is the hand's movements. In this system, a projected, virtual hand guides the user's (trainee) physical hand to perform the correct movements needed to complete the task. A trainer can provide real-time data to generate the virtual hand or previously record it. It can be used in different applications to guide the trainee toward the correct execution of a sequence of steps to complete an activity. An example of the projection hand can be seen in Figure 1.

2 ARCHITECTURE AND OVERVIEW

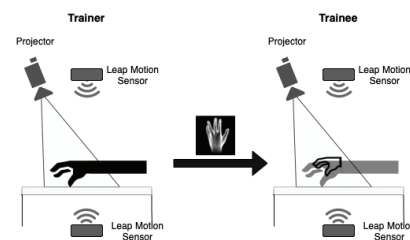


Figure 2: Physical structure diagram

2.1 Hardware Setup

The working area of the system is on top of glass table with 70 cm per 48 cm. The projector (RICOH PJ WXC1110) is positioned 46 cm above the working area and has a projection surface of 58 cm per 36 cm. Two sensors are used to obtain the hand data, one positioned above the projection surface (Ultraleap 3Di Stereo Hand Tracking Camera) and another under the glass table (Ultraleap Stereo IR 170). A diagram of the physical structure can be seen in Figure 2.

In training, the operation of different tools is necessary. However, during development, tools pose a problem because they can cause occlusion of the hand, and the sensor loses track of it. This situation can be mitigated by using two sensors, tracking the hand from different points of view and interpolating their data. An initial

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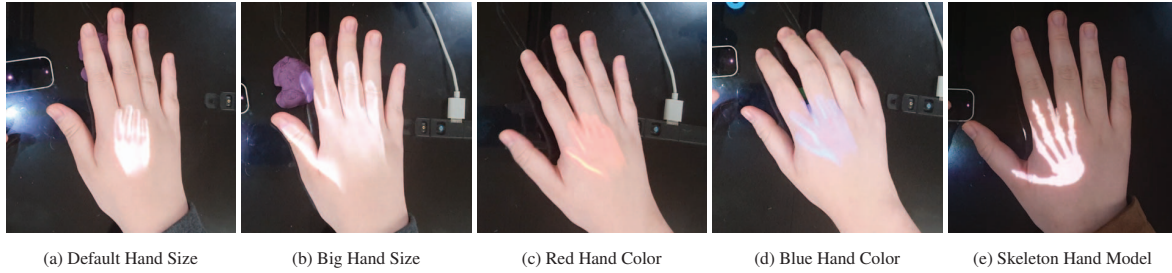


Figure 3: Different projected hand configurations

evaluation of the sensor capability was done by comparing only the upper sensor, only the bottom sensor, and both. The results can be seen in Table 1.

2.2 Operation Modes

This system can be used in two cases: real-time operation and playback operation. For real-time operation, the trainer and the trainee perform the task simultaneously, and the trainee receives a real-time projection of the trainer's operation. In this situation, a mirror setup is needed to capture both data simultaneously.

In the playback operation, the trainer's hand data is recorded and stored, and the visualization can be adapted with visual cues to improve the projection for the trainee. The trainee and the trainer can share the same setup in the playback operation at different times. For the trainer, the projector is used to calibrate and troubleshoot the data capture of the hand.

In both cases, the position and the rotation of the trainee's hand are tracked, allowing overlaying the pose of the trainer's hand that is being tracked or previously recorded by a separate sensor.

	Lower Sensor	Upper Sensor	Both
Nonoccluded	34.37%	84.44%	91.58%
Occluded by Pen	3.49%	66.51%	70.95%
Occluded by Mouse	5.37%	64.39%	89.44%
Occluded by Mallet	39.06%	64.71%	72.94%

Table 1: Hand Tracking Averages

2.3 Visual Effects

The hand projection can be modified accordingly to the task or the situation. Different hand models and appearances, such as different hand colors and sizes, are available. These modifications can be activated when needed, such as to call the trainee's attention or improve awareness of the operation.

Size Modification — The size can be changed to be smaller or bigger than the actual hand size, as shown in Figure 3a and 3b. This kind of modification can improve the detail of small movements or call the user's attention. The recommended size range varies from 0.3 to 2 times the actual hand size.

A smaller virtual hand creates a more projectable object on the surface of the physical hand. Therefore, depending on the application, the virtual hand is scaled from 0.3 to 0.7 of its original size.

Color Modification — The default color for the hand is white. It can be modified to different colors, such as green, red or blue, as shown in Figure 3c and 3d. The color can be used to indicate an object's color or as a warning for dangerous steps where the user needs to pay more attention.

Model Modification — Other types of hand models can be used, as shown in Figure 3e. For example, different textures can be applied, or the hand can be changed to a skeleton model.

Delay Modification — In the playback operation, delays can be introduced in the recording to facilitate the trainee's learning. These delays can create a resting period between steps to reduce the guiding speed of the trainee, facilitating the training for less experienced users.

2.4 Trainee Operation Features

The trainee has little control over the projection, it either happens automatically or under the trainer's control. However, the trainee can control the visualization of the trainer's hand by rotating their own hands, and the visualization will change its pitch, row, or yaw accordingly. It allows the trainee to view parts of the hand that could become occluded when the trainer moves.

3 DISCUSSION AND FUTURE WORK

We presented an AR system to support training scenarios using the projection of the hands. The preliminary results show the necessity of intervention techniques in the hand's visual to guarantee the user's attention. Increasing the hand size to attract the user's attention and the red color indicating danger are preferred techniques. Although the delay does not improve the experience, it can slow down the training to facilitate the trainee's session without compromising the trainer's performance, which can happen at full speed.

Future experiments will investigate the effectiveness of this system and compare the visualization modifications for different scenarios during a training scenario.

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